Practical 5

Aim: Study and Imp.ementation of K Mean Clustering

```
from sklearn.cluster import KMeans
import numpy as np
import matplotlib.pyplot as plt
# Generate some sample data
X = np.array([[1, 2], [1, 4], [1, 0], [4, 2], [4, 4], [4, 0]])
# Number of clusters
k = 2
# Create KMeans instance
kmeans = KMeans(n_clusters=k)
# Fit the model to the data
kmeans.fit(X)
# Get cluster centroids and labels
centroids = kmeans.cluster_centers_
labels = kmeans.labels_
# Plot the data points and centroids
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis', s=50, alpha=0.5, label='Data Points')
plt.scatter(centroids[:, 0], centroids[:, 1], c='red', marker='x', s=200, label='Centroids')
plt.title('K-means Clustering')
plt.xlabel('X')
plt.ylabel('Y')
plt.legend()
plt.show()
```